A Computational Framework for Clinical Trial Simulation of Glioblastoma Immunotherapy

Three-layer in silico pipeline: deterministic tumor-immune ODE core → stochastic virtual-patient cohort → survival analysis

a **INPUTS**

Model parameters

Anderson 2024 (ABC posterior);
11 parameters (mean + SD):

$\lambda_C, C_{max}, \eta, a_T, s_T, \rho$

$\varepsilon_C, r, d_T, s_M, d_M$

Initial conditions

$C_0 = 35{,}000$ cells

$T_0 = 100$ cells

$M_0 = 0$ cells

Treatment arms (4)

No Tx $(u_1 = 0, u_2 = 0)$

Anti-PD-1 $(u_1 = 0.7, u_2 = 0)$

CCR2 $(u_1 = 0, u_2 = 0.7)$

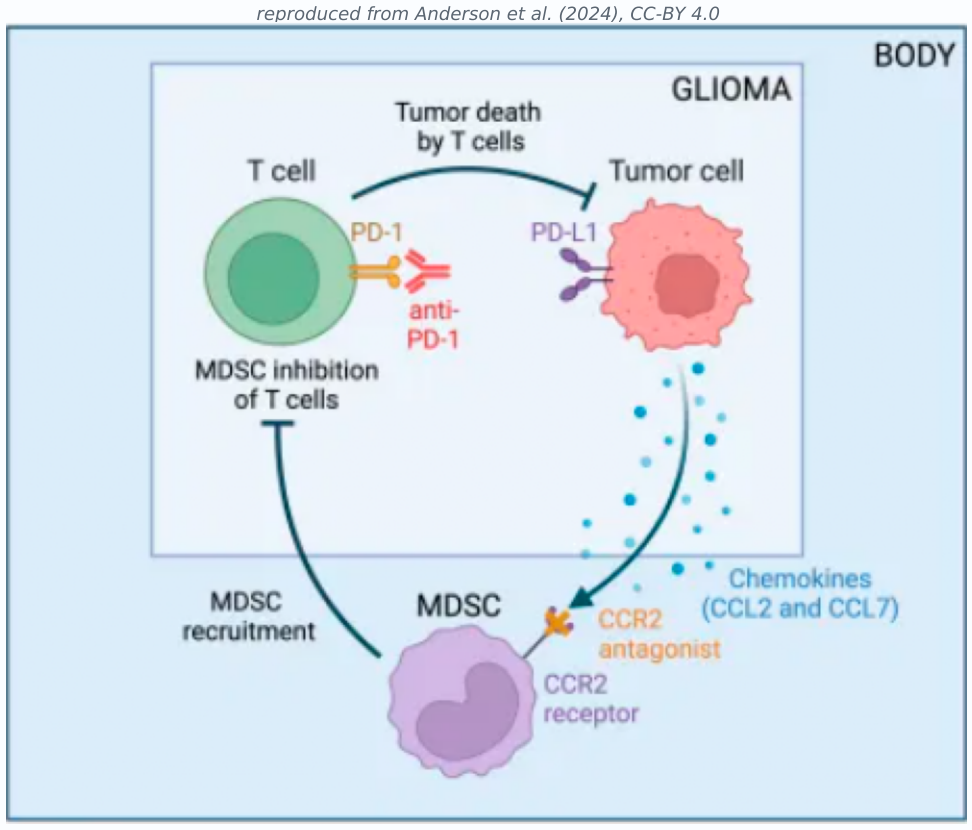
Combination $(0.7, 0.7)$

b **LAYER 1 — Deterministic core (ODE model)**

same parameters → same trajectory

Tumor-immune interactions in GBM under treatment

reproduced from Anderson et al. (2024), CC-BY 4.0



System of ordinary differential equations

$$\frac{dC}{dt} = \lambda_C C\left(1 - \frac{C}{C_{max}}\right) - \eta T C$$

$$\frac{dT}{dt} = \frac{a_T + s_T T C}{1 + \rho(1 - u_1) T (T + \varepsilon_C C)} - r T M - d_T T$$

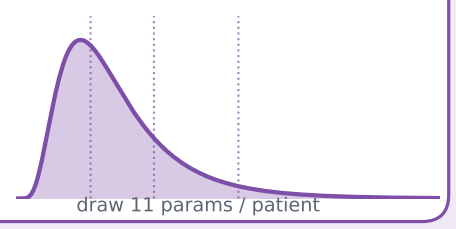
$$\frac{dM}{dt} = s_M (1 - u_2) C - d_M M$$

C tumor · T T cells · M MDSCs

numerical solvers used for per patient:
LSODA / RK45 / Radau benchmarked (event detection)

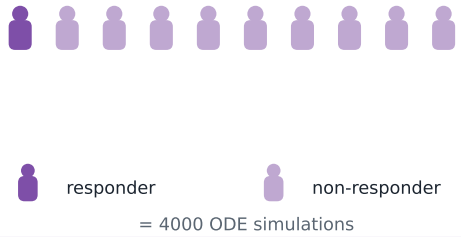
c **LAYER 2 — Stochastic layer (virtual-patient cohort, Monte Carlo)**

Log-normal sampling



draw 11 params / patient

1000 patients × 4 arms



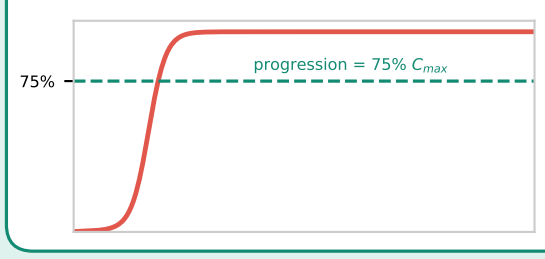
= 4000 ODE simulations

Clinical-realism layers

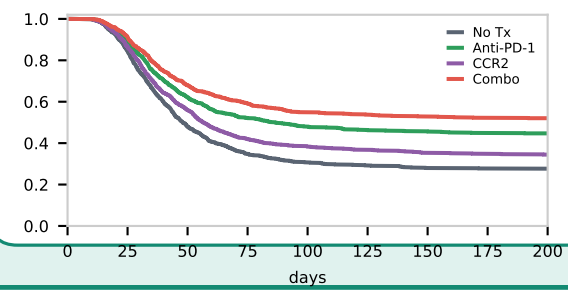
- Responder heterogeneity (70% / 30%, shown in cohort)
- Right-censoring — 15% dropout (median 90 d)
- Per-patient drug-effect scaling on u_1, u_2

d **LAYER 3 — Analysis layer (survival outcomes & solver benchmark)**

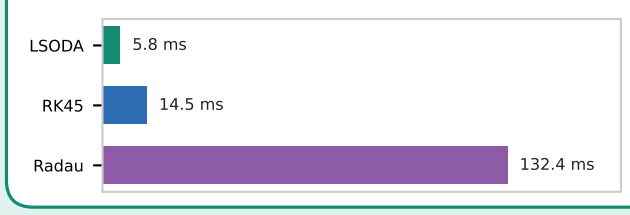
Event detection



Kaplan–Meier survival



Statistics & solver benchmark



Model: Anderson et al. (2024a, 2024b). Implementation: Python (scipy.integrate, scipy.optimize, lifelines). Endpoint: progression-free survival. Reproducible (seed = 42).